
Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper

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Abstract

Handheld grippers are increasingly used to collect human demonstrations due to their ease of deployment and versatility. However, most existing designs lack tactile sensing, despite the critical role of tactile feedback in precise manipulation. We present a portable, lightweight gripper with integrated tactile sensors that enables synchronized collection of visual and tactile data in diverse, real-world, and in-the-wild settings. Building on this hardware, we propose a cross-modal representation learning framework that integrates visual and tactile signals while preserving their distinct characteristics. The learning procedure allows the emergence of interpretable representations that consistently focus on contacting regions relevant for physical interactions. When used for downstream manipulation tasks, these representations enable more efficient and effective policy learning, supporting precise robotic manipulation based on multimodal feedback. We validate our approach on fine-grained tasks such as test tube insertion and pipette-based fluid transfer, demonstrating improved accuracy and robustness under external disturbances. Our project page is available at https://binghao-huang.github.io/touch_in_the_wild/.

Keywords: Visuo-Tactile Sensing, Fine-Grained Manipulation, Scalable Data Collection

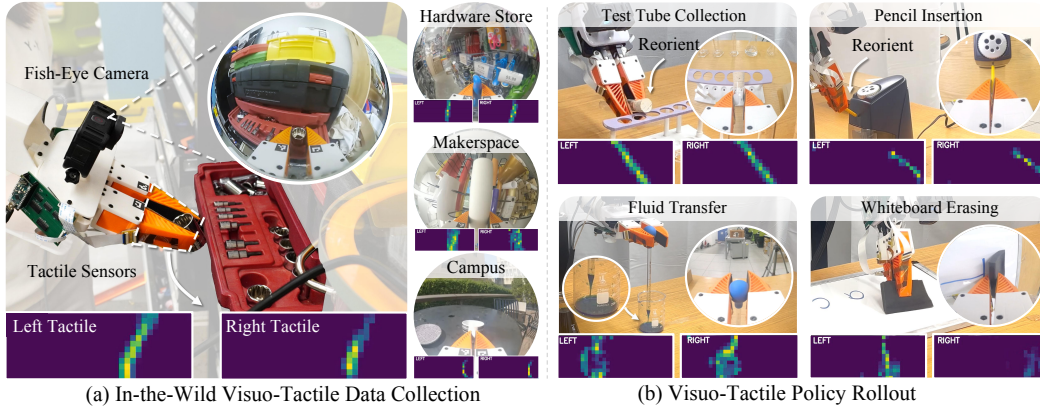


Figure 1: (a) Our portable handheld gripper enables synchronized collection of visual and tactile data through integrated multimodal sensing, supporting large-scale data collection in the wild. (b) We introduce a multimodal representation learning framework that fuses visual and tactile inputs to support fine-grained downstream manipulation tasks.

1 Introduction

Humans naturally rely on both vision and touch when interacting with the physical world. Whether inserting a key into a lock or adjusting a pipette during a lab experiment, tactile feedback plays a

critical role in guiding precise motor actions, especially in situations where visual information may be unreliable due to occlusion, poor lighting, or dynamic backgrounds. While vision provides global, semantic context, touch offers local, high-resolution feedback about contact and force. The integration of these two modalities is fundamental to effective manipulation in everyday environments.

Recent advances in handheld grippers have made it easier to collect human demonstrations “in the wild.” However, most existing systems focus exclusively on visual sensing, largely neglecting tactile feedback. This gap limits their usefulness for capturing the fine-grained, contact-rich strategies humans use in real-world tasks. Moreover, relying on vision alone makes these systems vulnerable to environmental variability, whereas tactile sensing offers a complementary and robust signal that is invariant to lighting conditions and camera viewpoint.

Two key challenges have prevented widespread visuo-tactile data collection in the wild: (i) *Portable tactile hardware*. Most existing tactile sensors are bulky, rigid, or not robust enough. For example, soft bubble sensors [1] are too large for handheld use, while other optical-based sensors [2] are often bulky or less robust for long-term in-the-wild use, making them less suitable for mobile or outdoor deployment. (ii) *Learning from multimodal data*. Tactile and visual signals differ significantly in scale and in the nature of collected information. Tactile inputs are local and physical, while vision encodes broader spatial context. Learning effective representations that integrate both modalities—particularly from large-scale, unstructured datasets—remains an open challenge.

To address these issues, we present a Portable Visuo-Tactile System for large-scale data collection and multimodal policy learning in real-world settings. Our contributions are threefold: (1) *A lightweight, handheld visuo-tactile gripper*. We integrate flexible piezoresistive tactile sensors into a soft, handheld gripper to enable portable, visuo-tactile, in-the-wild data collection. The system captures human manipulation demo trajectories across a wide range of environments, both indoors and outdoors. (2) *A multimodal representation and policy learning framework*. We introduce a masked autoencoding approach that uses cross-attention to jointly learn from visual and tactile inputs while preserving modality-specific characteristics. This design enables policies to interpret fine-grained tactile feedback more effectively, leading to improved sample efficiency and manipulation accuracy. (3) *A in-the-wild multimodal manipulation dataset*. We curated a diverse dataset of over 2.6 million visuo-tactile pairs, comprising more than 2,700 demonstrations across 43 manipulation tasks in 12 indoor and outdoor environments. This dataset supports effective visuo-tactile pretraining, and the resulting encoder significantly improves downstream policy learning. We are committed to open-sourcing the dataset.

We validate our system on fine-grained robotic manipulation tasks in the real world, such as test tube insertion and pipette-based fluid transfer, demonstrating successful policy transfer and robustness to environmental disturbances. Our results underscore the promise of portable visuo-tactile platforms in bridging the gap between human demonstrations and robot learning in complex, real-world settings.

2 Related Works

Scalable multi-sensory data and learning. Reinforcement and imitation learning have become central to robotic manipulation [3–9], but progress is often bottlenecked by the lack of tactile sensing or access to large-scale, high-quality visuo-tactile datasets. While simulation can mitigate data scarcity, simulated tactile signals tend to diverge significantly from their real-world counterparts, limiting transferability [5, 10–12]. This makes scalable real-world visuo-tactile data collection increasingly important. However, acquiring multi-sensory data at scale remains challenging. Beyond RGB streaming, each additional modality [13–17]—such as audio [18], force, or tactile—introduces added hardware complexity, synchronization overhead, and environmental constraints. As a result, most prior efforts have limited data collection to structured indoor environments [19].

However, tactile sensing is most valuable in uncontrolled, in-the-wild settings, where vision may degrade due to poor lighting or background clutter [20, 21], while contact forces remain stable. Prior “in-the-wild” systems have demonstrated strong visual-only performance on simple tasks, but they overlook the complementary benefits of touch [22, 23]. To bridge this gap, we propose a portable handheld visuo-tactile system that combines a fisheye RGB camera with a lightweight, flexible

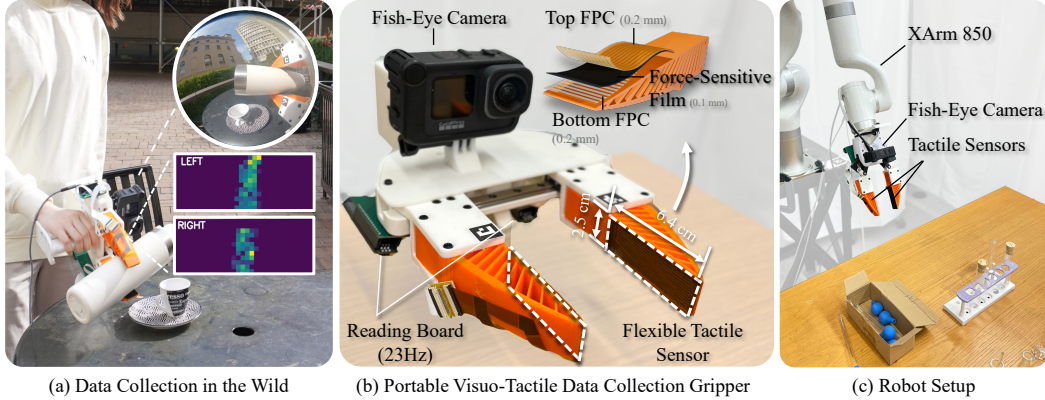


Figure 2: (a) Multimodal data collection in the wild using our portable visuo-tactile system, with example tactile signals from both fingers. (b) Close-up of the handheld gripper, equipped with flexible tactile sensors and a fisheye camera for synchronized visual-tactile capture. (c) Robotic setup for downstream tasks, featuring an XArm 850 with the same sensor configuration.

tactile array. This setup enables synchronized, large-scale collection of visual and tactile data across diverse, unstructured environments. Leveraging this system, we build a visuo-tactile dataset and train a masked reconstruction encoder that (i) accurately recovers tactile signals, and (ii) enhances downstream policy learning through visuo-tactile representations. In contrast to previous methods that directly merge vision and touch in a unified 3D space [19], our approach enables scalable cross-modal representation learning, fostering stronger correlations between visual and tactile modalities for fine-grained manipulation.

Visuo-tactile manipulation. Tactile feedback plays a critical role in human manipulation, particularly when visual input is occluded or ambiguous [24, 25]. Similarly, tactile sensing enables more adaptive robotic manipulation for tasks that are complex or require precision. As a result, there is growing interest in integrating vision and touch to improve robotic manipulation [9, 19, 26–29, 29–35]. Much of this prior work relies on optical tactile sensors, which image surface deformations to infer contact geometry and texture [15, 31, 36–40]. While these sensors provide rich signals, they are typically rigid, bulky, and less robust for long-term in-the-wild use, limiting their applicability in portable or unstructured settings.

In contrast, we use thin, flexible tactile sensors [19, 41, 42] that directly measure force distributions over the contact surface. Embedded in soft robotic fingers, these sensors provide consistent, object-agnostic representations that are easier to generalize and better suited for scalable, real-world learning. Although such sensors have been explored in lab setting [19, 43], large-scale, in-the-wild visuo-tactile datasets with synchronized RGB and tactile data capturing physical interactions have not previously been available. Our work fills this gap by collecting—and publicly releasing—a diverse, real-world visuo-tactile dataset. It spans a wide range of tasks and environments, laying the foundation for scalable multimodal learning and robust policy development. We acknowledge the concurrent work ViTaMIn [44], which also proposes a portable visuo-tactile data collection system and a cross-modal learning framework for contact-rich manipulation. Both of our work shares high-level goals, but the system design, representation learning approach, and task emphasis differ in several key aspects. We appreciate their contribution to this emerging area and encourage the reader to refer to their work for complementary insights.

3 Visuo-Tactile Data Collection System

3.1 Scalable Flexible Tactile Sensors

As shown in Fig. 2 (a), we embed thin, matrix-based tactile pads into the soft, fin-shaped fingers of our handheld gripper. The sensor architecture builds on the triple-layer design from 3D-ViTac [19], adapted to fit the geometry of the adaptive fin-shaped gripper [22]. Each tactile pad consists of a piezoresistive sensing layer sandwiched between two flexible printed circuits (FPCs). To accommodate the elongated, flexible fins, we introduce two key modifications: (1) *Higher spatial*

resolution. Capturing contact patterns along the finger’s length requires denser spatial sampling. The stainless-steel electrodes used in 3D-ViTac limit both resolution and signal stability. By replacing them with FPC electrodes, we achieve uniform trace pitch, improved robustness, and a per-pad resolution of 12×32 taxels, each measuring a $2 \times 2 \text{ mm}^2$ area. This allows us to capture fine-grained, dynamic contact signals. (2) *Rapid, scalable fabrication.* The use of FPCs enables tool-free assembly. Each pad can be fabricated in under five minutes and mounted on the gripper in an additional two, supporting scalable deployment for large-scale tactile data collection.

3.2 Portable Multi-Modal Sensing System

In-the-wild large-scale data collection. To enable real-world visuo-tactile data collection at scale, we design a compact and ergonomic handheld gripper that integrates both sensing modalities. Each tactile pad connects to a custom Arduino-based PCB, with two boards neatly housed beneath the gripper’s palm (Fig. 2 (b)). The full handheld unit—including batteries—weighs approximately 962 g, making it comfortable for prolonged use.

At the firmware level, we optimize the serial protocol to stream each 12×32 pad at 23 Hz and synchronize with the visual information from the fish-eye camera. Tactile frames are timestamped directly on the microcontroller and transmitted over USB to a host device (e.g., a laptop or any portable Ubuntu system). The battery-powered, handbag-sized system is easily deployed in grocery stores, outdoor markets, and other unstructured, in-the-world environments (Fig. 2 (a)), enabling high-throughput and scalable visuo-tactile data collection.

Multi-modal data synchronization. Precise alignment between vision and touch is essential for learning effective visuo-tactile representations. Although both the tactile system and the GoPro camera operate above 20 Hz (e.g., tactile sensors at 23 Hz, GoPro 9 at 60 Hz), aligning their data streams poses challenges due to clock drift and limited timestamp precision on the camera.

We address this with a hardware-free synchronization strategy: (i) *Video stream.* Before each demonstration, a QR code displaying the current host time is shown to the camera, refreshed at 30 Hz. (ii) *Tactile stream.* Tactile data is published via ROS2 at 23 Hz, with each packet carrying a host-clock timestamp. (iii) *Post-processing.* During offline processing, we decode the QR code sequence from the video, recover exact host timestamps for each frame, and align them with the tactile data using the shared clock reference. This procedure yields tightly aligned visual and tactile recordings—without the need for external synchronization hardware—enabling accurate multimodal supervision for downstream learning.

4 Visuo-Tactile Representation and Policy Learning

To perform precise manipulation, robots must effectively integrate both visual and tactile signals. RGB images provide global, semantic context—such as object identity and workspace layout—while tactile signals offer local, contact-rich feedback that is often occluded from vision [45]. These complementary modalities follow different statistical distributions, making it non-trivial to learn unified representations that preserve modality-specific information while enabling effective cross-modal reasoning. We propose a two-stage learning framework (Fig. 3) that first learns a joint visuo-tactile representation via masked tactile reconstruction, and then integrates the learned representation into a diffusion policy [46] for downstream manipulation tasks.

4.1 Problem Formulation

Let $\mathcal{D}_{\text{pretrain}} = \{(I, T)\}$ be a large-scale dataset of synchronized RGB-tactile frame pairs, where $I \in \mathbb{R}^{3 \times 224 \times 224}$ is an RGB image captured from a wrist-mounted fisheye camera, and $T \in \mathbb{R}^{1 \times 24 \times 32}$ is a tactile image composed of vertically stacked fingertip sensor readings. The goal is to learn a multimodal encoder E_ϕ , trained via self-supervised learning, that fuses these two modalities into a joint representation $z_{\text{fusion}} = E_\phi(I, T)$ that preserves modality-specific structure to support downstream manipulation tasks.

We divide the learning into two stages: *Stage 1:* Pretrain E_ϕ using a masked autoencoding objective that reconstructs full tactile images from partially observed tactile input and corresponding visual

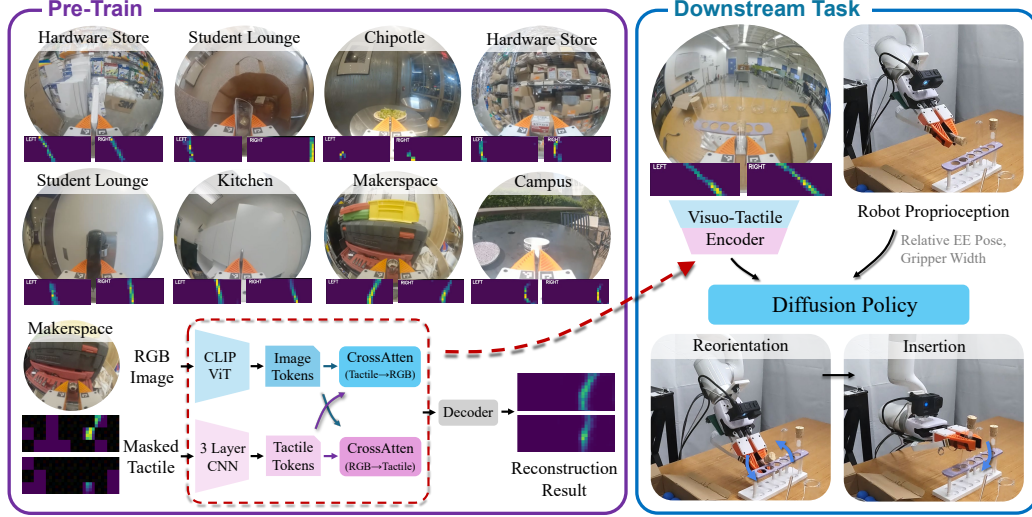


Figure 3: **Method Overview** of our two-stage pipeline. *Stage 1*: We pretrain a visuo-tactile encoder via cross-modal reconstruction using a large-scale dataset collected across diverse indoor and outdoor environments. *Stage 2*: The pretrained encoder is combined with robot proprioception to condition a diffusion policy for downstream tasks such as object reorientation and insertion.

frames. *Stage 2*: Use the pretrained encoder within a diffusion policy to learn manipulation behaviors from human demonstrations collected using our portable visuo-tactile gripper.

4.2 Stage 1: Visuo-Tactile Representation Learning

While prior work often relies on contrastive learning to align embeddings from different modalities [44, 47, 48], such objectives tend to suppress the fine-grained, geometry-sensitive signals captured by tactile sensors. Instead, we adopt a *masked autoencoding objective* [49], which reconstructs missing tactile regions conditioned on partially observed tactile input and visual context. This formulation encourages the encoder to retain tactile-specific information while leveraging vision for inference.

Formally, we jointly optimize the encoder E_ϕ and decoder D_ψ via

$$(\phi^*, \psi^*) = \arg \min_{\phi, \psi} \mathbb{E}_{(I, T) \sim \mathcal{D}_{\text{pretrain}}} \left\| T - D_\psi(E_\phi(I, T)) \right\|_2^2, \quad (1)$$

where E_ϕ is our visuo-tactile encoder and D_ψ is the tactile reconstruction decoder.

Tactile encoder. Each tactile reading consists of two fingertip arrays, each of shape $1 \times 12 \times 32$, stacked vertically to form a $1 \times 24 \times 32$ tactile image. We then apply a fixed colormap to convert this single-channel map into a 3-channel RGB tactile image. This image is divided into non-overlapping 4×4 patches, resulting in a 6×8 patch grid. During training, we randomly mask 60–80% of the patches in 95% of samples using a learnable token T_{mask} ; the remaining 5% are shown in full. The masked tactile input is defined as:

$$T_{\text{visible}} = M \odot T + (1 - M) \odot T_{\text{mask}}, \quad (2)$$

where $M \in \{0, 1\}^{6 \times 8}$ is a binary patch mask. The visible tactile input T_{visible} is processed by a 3-layer CNN to produce a d -dimensional embedding z_{tac} , where $d = 768$.

Vision encoder. The RGB image I is processed by a ViT-B/16 encoder initialized from CLIP [50]. We fine-tune all layers with a learning rate of 3×10^{-5} , and extract the final [CLS] token as the d -dimensional visual embedding z_{img} .

Cross-modal fusion. To integrate the tactile and visual features, we apply two rounds of multi-head cross-attention (MHAttn):

$$z'_{\text{tac}} = \text{MHAttn}(Q = z_{\text{tac}}, K = z_{\text{img}}, V = z_{\text{img}}) \xrightarrow{\text{LayerNorm}} z''_{\text{tac}}, \quad (3)$$

$$z'_{\text{img}} = \text{MHAttn}(Q = z_{\text{img}}, K = z'_{\text{tac}}, V = z'_{\text{tac}}) \xrightarrow{\text{LayerNorm}} z''_{\text{img}}. \quad (4)$$

We concatenate the updated embeddings to obtain the fused representation:

$$z_{\text{fusion}} = [z''_{\text{tac}}; z''_{\text{img}}] \in \mathbb{R}^{2d}. \quad (5)$$

Tactile reconstruction decoder. The fused feature z_{fusion} is passed through a two-layer MLP followed by a sigmoid activation to produce the reconstructed tactile image $\hat{T} \in \mathbb{R}^{1 \times 24 \times 32}$, i.e.,

$$\hat{T} = D_{\psi}(z_{\text{fusion}})$$

We use a full-image reconstruction loss:

$$\mathcal{L}_{\text{stage1}}(\phi, \psi) = \|T - \hat{T}\|_2^2,$$

which encourages both local contact inference and global structural understanding.

Stabilization via EMA. We maintain a target encoder updated via exponential moving average (EMA) of the online weights with a decay factor of 0.9995. The tactile CNN and cross-attention layers are optimized with a learning rate of 1×10^{-4} , while the CLIP backbone is fine-tuned with 3×10^{-5} . The EMA encoder is used for checkpointing and attention visualization.

4.3 Stage 2: Policy Learning via Behavior Cloning

Once the visuo-tactile encoder E_{ϕ} is pretrained, it is integrated into a conditional diffusion policy for downstream manipulation tasks.

Observation space. At each timestep t , the robot first receives raw sensory inputs

$$(I_t, T_t, p_t),$$

where I_t and T_t are the RGB and tactile images, and p_t denotes the proprioceptive state (e.g., end-effector pose, gripper width). I_t and T_t are then passed through the pretrained encoder to produce the visuo-tactile embedding $z_t = E_{\phi}(I_t, T_t)$. The diffusion policy is then conditioned on

$$o_t = (z_t, p_t),$$

i.e. the fused visuo-tactile embedding together with proprioception.

Diffusion policy. We adopt a conditional diffusion policy [46]. Instead of regressing actions directly, we train a noise predictor:

$$\hat{\epsilon}_t^k = \epsilon_{\theta}(a_t^k, o_t, k), \quad (6)$$

where a_t^k is the noisy action at diffusion step k , o_t is the full conditioning observations, and $\hat{\epsilon}_t^k$ is the predicted noise. During training, we sample k uniformly and add Gaussian noise ϵ_t^k to the ground-truth action a_t^0 . We formulate our loss as the MSE loss between added and predicted noise:

$$\mathcal{L}_{\text{stage2}} = \mathbb{E}_{t,k} [\|\epsilon_t^k - \hat{\epsilon}_t^k\|_2^2]. \quad (7)$$

At inference, we start from $a_t^K \sim \mathcal{N}(0, I)$ and denoise for K steps:

$$a_t^{k-1} = \alpha(a_t^k - \gamma \epsilon_{\theta}(a_t^k, o_t, k)) + \mathcal{N}(0, \sigma^2 I), \quad (8)$$

where α, γ, σ are hyperparameters.

Training details. We use Diffusion Policy’s convolutional U-Net [51] with DDIM inference [52]. The model is conditioned on the fused visuo-tactile embedding and two consecutive proprioceptive observations. All encoder components—including the CLIP backbone, tactile CNN, and cross-attention layers—are fine-tuned during this stage using a learning rate of 3×10^{-5} .

5 Experiments

In this section, we address several key questions regarding the role of touch in fine-grained manipulation and the impact of different pretraining strategies on downstream tasks. Specifically, we investigate: (1) How does touch enable fine-grained manipulation? (2) How does pre-trained visuo-tactile encoder aid robust policy learning? (3) How do variations in pretraining affect downstream task performance?

5.1 Large-Scale Visuo-Tactile Data and Pre-Training

To enable effective visuo-tactile pretraining, we curated a diverse dataset of over 2.6 million visuo-tactile pairs collected from 12 indoor and outdoor environments. This dataset comprises more than 2,700 demonstrations and spans 43 manipulation tasks. We categorize the data into three groups: (1) the four core tasks presented in this paper, (2) other indoor tasks that enhance distributional diversity, and (3) over 30 in-the-wild tasks designed to capture complex, real-world scenarios. The distribution of demonstrations across these three categories is shown in Figure 4.

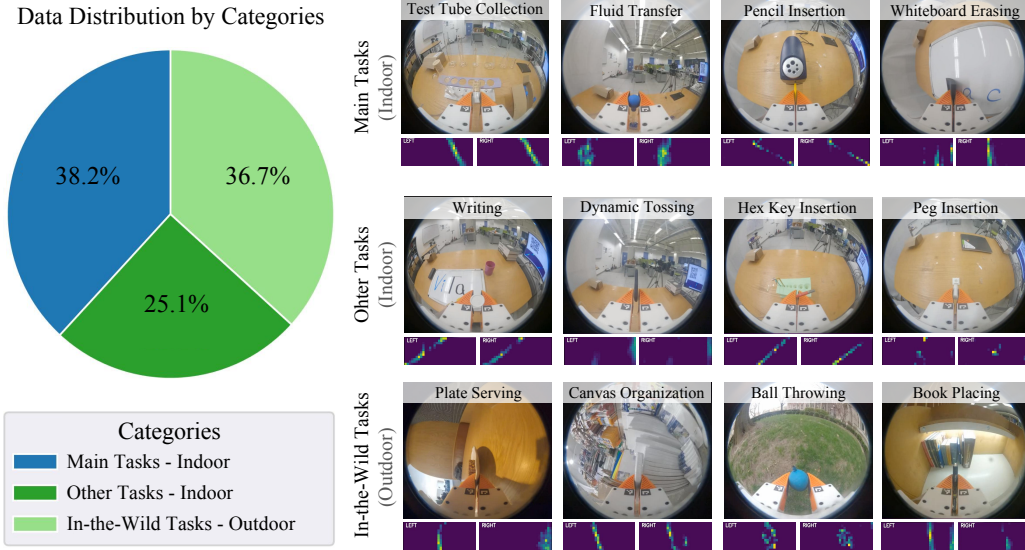


Figure 4: **Pretraining Data Distribution.** Our dataset comprises over 2,700 demonstrations, evenly split across three categories: (1) the four core tasks introduced in this paper, (2) other indoor tasks to broaden the data distribution, and (3) in-the-wild tasks collected in diverse outdoor environments. We include representative examples from each category to highlight the variety in both task complexity and environmental context.

We evaluate the quality of the learned representations from our pretrained visuo-tactile encoder through two complementary qualitative analyses. First, we assess cross-modal reconstruction by providing the encoder with partially masked tactile and RGB images, then measuring its ability to reconstruct the missing tactile inputs. This evaluates whether the model has learned meaningful visuo-tactile associations and can infer tactile information from visual cues, both in in-distribution (seen during training) and out-of-distribution (unseen during training) environments. Second, we examine the encoder’s attention mechanisms by visualizing self-attention maps from the final layer of the Vision Transformer (ViT). This analysis reveals whether the model consistently attends to relevant contact regions in RGB images, even across varied and unstructured scenarios. Figure 5 illustrates qualitative results from four representative tasks: two from in-distribution and two from out-of-distribution test data.

We further analyze how the amount of pretraining data affects reconstruction performance and representation quality. Figure 6 shows training loss curves across different dataset sizes, demonstrating that larger datasets result in lower reconstruction losses and more stable training. To illustrate this effect, we visualize a representative reconstructed tactile image alongside its corresponding attention heatmap. As the dataset size increases, the attention maps become more sharply focused on the gripper contact regions, and the reconstructed tactile images exhibit reduced noise and clearer structure—both indicating improved cross-modal representation learning.

5.2 Experiments Setup

We evaluate our multi-modal sensing and learning system on four challenging real-world robotic tasks. Below are the basic descriptions and evaluation metrics for all tasks:

(1) *Task Requiring In-Hand State Information*

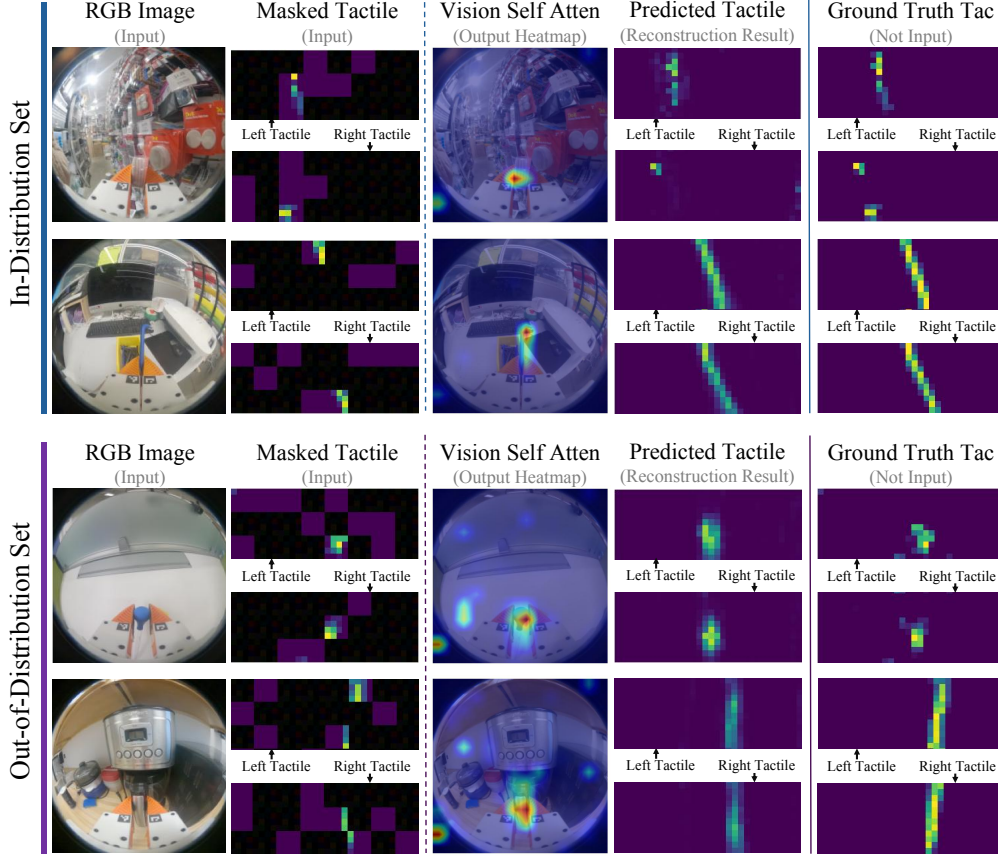


Figure 5: **Qualitative Results of Pre-Training.** We present four examples showcasing the results of our visuo-tactile pretrained encoder, highlighting both tactile image reconstruction and ViT self-attention heatmaps. For tactile image reconstruction, the encoder successfully reconstructs tactile images for both in-distribution and out-of-distribution inputs. Additionally, we observe that the vision encoder consistently focuses on the gripper contact region, regardless of the background or whether the object is seen or unseen.

Test Tube Collection. The robot must pick up a test tube from a box, reorient it in-hand using the test tube rack, and precisely insert it into the test tube rack. *Evaluation Metric:* The task is considered successful if the test tube is fully inserted into the test tube rack without being dropped or broken.

Pencil Insertion. The robot needs to insert a pencil into a sharpener. Since the pencil is initially grasped upright, the robot must first reorient it before performing a precise insertion. *Evaluation Metric:* The task is considered successful if the pencil is accurately inserted into the sharpener.

(2) Task Requiring Fine-Grained Force Information

Fluid Transfer. The robot uses a pipette to transfer water between containers. It must grasp the pipette firmly, apply just enough pressure to extract liquid without dropping it. Then the robot need to move to the top of other container and gently squeeze to release the water into it. This task demands continuous and sensitive force modulation. *Evaluation Metric:* The task is considered successful if the water is transferred into the second container without spilling.

Whiteboard Erasing. The robot uses a soft eraser to remove two strokes of text from the whiteboard. It must apply the right amount of pressure to erase the marker ink without exceeding force limits that could damage the system. The task requires consistent and controlled force application throughout. *Evaluation Metric:* The task is considered successful if all visible marker ink is removed from the whiteboard.

In the experiments, we compare our methods with the following baselines.

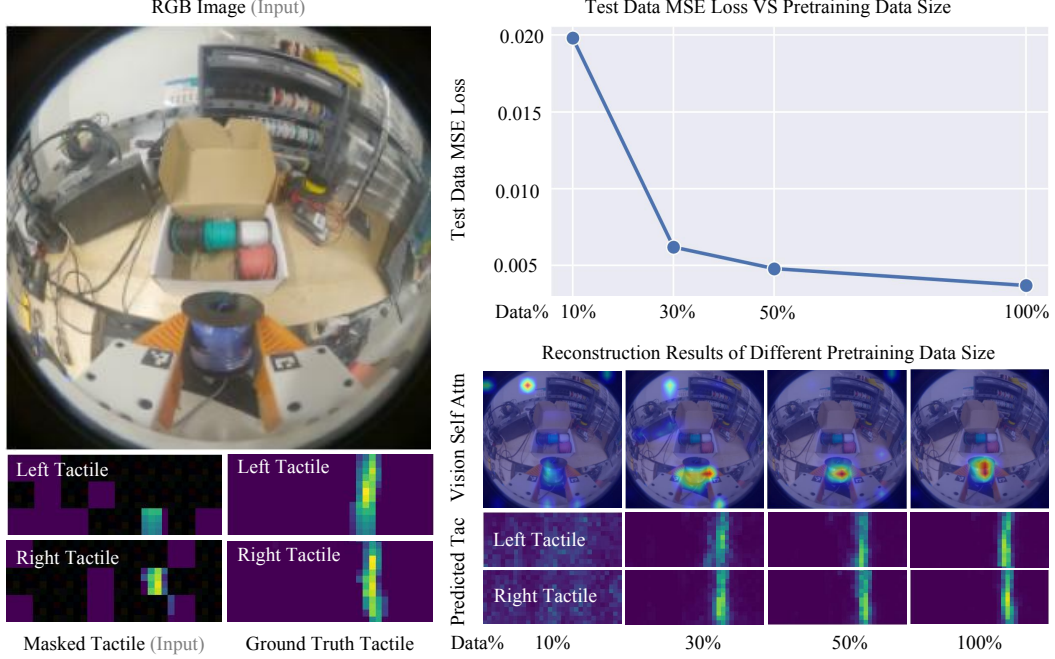


Figure 6: **Impact of pretraining dataset size on cross-modal reconstruction.** On the left, we show a representative visuo-tactile example with RGB image, the masked tactile input, and the ground-truth tactile image. On the right, the top panel plots test data MSE loss against the amount of pretraining demonstrations, and the bottom panel displays reconstructed tactile images alongside their visual self-attention heatmaps for four different dataset amounts. As the amount of pretraining data grows, the test MSE steadily decreases and the reconstructions become notably cleaner and more accurate.

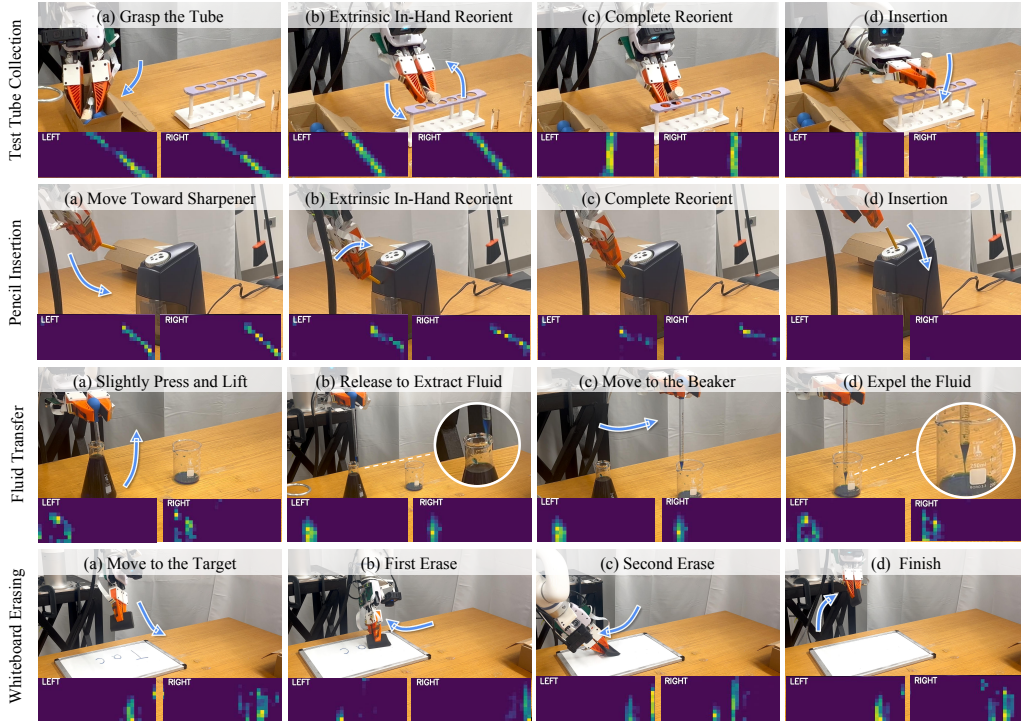


Figure 7: **Quantitative Results.** We evaluate our visuo-tactile policy across four fine-grained manipulation tasks. Detailed descriptions and metrics can be found in Sec. 5

- (1) *Vision-Only*. This method feeds one RGB image as input to CLIP, and extracts a 768-dimensional CLIP embedding. This embedding, along with the robot’s proprioceptive information, is then fed into the image-based diffusion policy. We follow the same implementation as outlined in Chi et al. [22].
- (2) *Ours w/o Cross Attention*. This method does not employ any cross-attention between vision and touch. Instead, it processes two tactile images (from the left and right fingers) through a 3-layer CNN to produce a 512-dimensional feature vector, which is then simply concatenated with the visual embedding and proprioceptive states as input to the diffusion policy.
- (3) *Ours w/o Pretraining*. This method uses the visuo-tactile encoder proposed in the paper, with the vision backbone initialized from CLIP and the other parts of the joint encoder initialized from scratch (no visuo-tactile pretraining). The resulting embedding is concatenated with proprioceptive inputs to condition the diffusion policy.
- (4) *Ours w/ Pretraining*. This method uses the visuo-tactile encoder proposed in the paper, with weights obtained from visuo-tactile pretraining. Similarly, the output embedding from the visuo-tactile encoder is concatenated with robot proprioception as conditioning for the diffusion policy.

For each of the four tasks, we conduct 20 trials with moderate randomization of the initial robot position and environmental conditions. All policies are trained for 60 epochs, at which point they have converged. The results are presented in Table 1.

Modalities	Tasks Requiring In-Hand State Information						
	Grasp	Test Tube Collection			Pencil Insertion		
		Reorient	Insert	Whole Task	Reorient	Insert	Whole Task
Vision-Only	1.00	0.25	0.25	0.25	0.50	0.65	0.45
Ours w/o Cross Attention	1.00	1.00	0.50	0.50	0.80	0.75	0.70
Ours w/o Pretraining	1.00	1.00	0.70	0.70	0.80	0.80	0.75
Ours w/ Pretraining	1.00	1.00	0.85	0.85	0.95	0.90	0.85

Modalities	Tasks Requiring Fine-Grained Force Information						
	Fluid Transfer				Whiteboard Erasing		
	Acquire	Transfer	Expel	Whole Task	First Erase	Second Erase	Whole Task
Vision-Only	0.95	0.85	0.55	0.55	0.65	0.65	0.55
Ours w/o Cross Attention	0.90	0.75	0.75	0.70	0.70	0.50	0.45
Ours w/o Pretraining	1.00	0.90	0.80	0.80	0.60	0.75	0.60
Ours w/ Pretraining	1.00	1.00	0.90	0.90	0.70	0.75	0.70

Table 1: **Comparison with Baselines.** We evaluate our policy over 20 episodes and the best performance for each task is bolded.

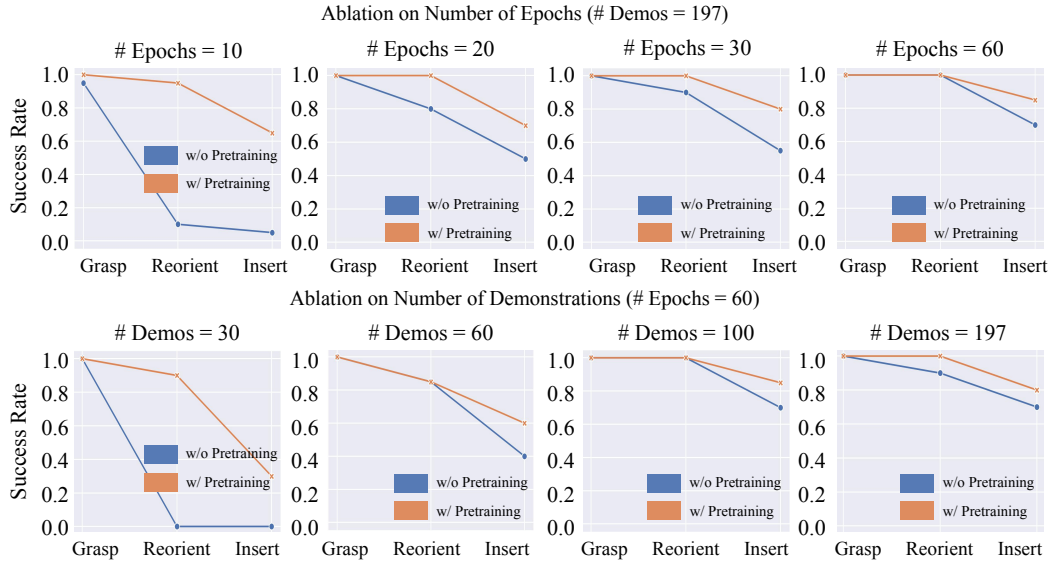


Figure 8: Ablation results for varying numbers of demonstrations and epochs for Test Tube Collection Task. Our findings show that the policy with visuo-tactile pretraining consistently outperforms the policy without pretraining, both in low-epoch and low-demonstration regimes.

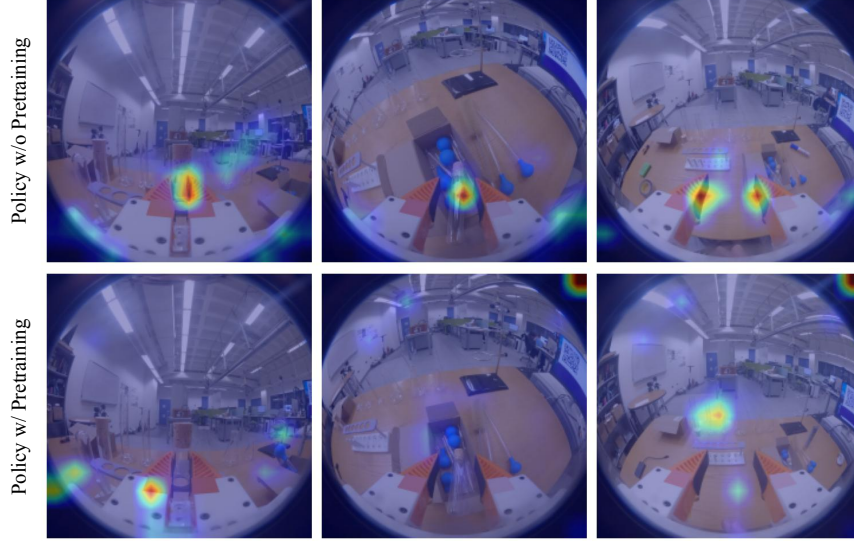


Figure 9: **Visual Transformer attention maps after downstream fine-tuning.** In Visual Transformer’s attention maps, policies with pretraining tend to focus on task-relevant contact regions, such as the gripper and the object. In contrast, policies without pretraining often attend to background elements, like table edges or box boundaries, as a shortcut to learning the task, which results in less stable performance during downstream tasks.

5.3 Analysis and Discussion

Our system enhances a handheld gripper by integrating tactile sensing and training a large-scale visuo-tactile encoder to further improve manipulation policies. We observe three key benefits from incorporating touch and leveraging pretrained representations.

(1) *Tactile feedback provides explicit in-hand state information.* In a single-camera handheld setup, visual inputs often suffer from severe occlusions. For example, in the test tube insertion task, the vision-only policy relied heavily on the color of the wooden cork to infer orientation. A minor change (such as switching to a lighter cork with less distinct features) confused the vision policy and degraded performance in reorientation. The tactile policy, however, remained unaffected by such variations.

(2) *Tactile feedback improves detection of critical state transitions.* In fine-grained force-controlled tasks like fluid transfer, accurately identifying when one action phase ends and another begins is essential. The vision-only policy struggles with this because visual features—such as the gripper’s width—remain similar before and after contact, making it difficult to determine if the gripper is still squeezing or has completed the action. As a result, the policy often skips the “expel water” phase prematurely, incorrectly assuming the transfer is finished. In contrast, the tactile policy receives direct feedback from pressure changes, enabling it to detect subtle shifts in force and correctly infer when the water has been fully expelled. The tactile feedback helps the policy transition smoothly between task stages, improving both accuracy and robustness.

(3) *Joint visuo-tactile encoders enable more coordinated use of vision and touch.* Naive fusion approaches that simply concatenate visual and tactile features, such as the policy without cross attention, often fail to meaningfully combine the two modalities. As a result, the policy may over-rely on one input while neglecting the other. This imbalance was evident in the whiteboard erasing task: the policy without cross attention applied excessive force to maximize tactile signal changes, which triggered safety warnings and caused the task to fail. In contrast, our jointly trained visuo-tactile encoder learns to coordinate both modalities, allowing the policy to modulate force more appropriately based on visual context and tactile feedback. This balanced integration reduces failure cases caused by overuse or misuse of either modality.

Pretraining Ablations: Varying Number of Demonstrations and Epochs.

To evaluate the effectiveness of pretraining, we assess the performance of the Test Tube Collection task under varying numbers of demos and epochs. The results are shown in Figure 8. We found that pretraining provides significant benefits, particularly in low-data and low-epoch training settings.

(1) *Low-Data Regime (Fewer Than 60 Demonstrations)*. When training with only 30 or 60 demonstrations, the policy without pretraining often becomes stuck after grasping, uncertain how to proceed. In contrast, the policy with pretraining—even with just 30 demonstrations—follows much smoother trajectories and usually only fails during the final insertion step. We believe that pretraining helps the joint encoder learn visuo-tactile correlations early on, which enables the downstream policy to focus on learning effective action trajectories.

(2) *Low-Epoch Regime (Fewer Than 60 Epochs)*. In the low-epoch regime, we observed that the policy without pretraining was more sensitive to initial environmental configurations. For instance, when the test tube was placed at a steep incline and was difficult to grasp, an imperfect grasp position had a considerable impact on the execution of the reorientation task. The policy without pretraining sometimes over- or under-reoriented, resulting in failure. We believe that in the low-epoch regime, the policy with pretraining benefits from prior knowledge that emphasizes tactile cues, which makes it more robust to environmental disturbances.

To further understand the performance gap between the policies with and without pretraining observed during real-world evaluation, we analyzed the attention maps from the ViT module in the Test Tube Collection task, as shown in Figure 9. The policy with pretraining consistently focuses attention on the gripper-object contact region, suggesting that it retains spatial priors critical for manipulation—priors learned during large-scale visuo-tactile pretraining. In contrast, the policy without pretraining frequently directs attention to irrelevant background features, such as table edges or nearby objects, likely in an attempt to infer contact indirectly. This misplaced focus can degrade performance, particularly in cluttered or visually diverse environments. These attention patterns provide insight into why the policy with pretraining not only learns more efficiently but also generalizes better across task variations. Additional analysis of loss convergence for both policies is provided in the appendix.

6 Conclusion and Limitations

In this work, we present a handheld gripper enhanced with tactile sensing and introduce a large-scale visuo-tactile dataset. We demonstrate its utility by pretraining a visuo-tactile joint encoder and evaluating it on several fine-grained manipulation tasks using a single-arm robot equipped with a parallel gripper, mirroring the handheld setup. Looking ahead, we aim to extend this approach to multi-finger dexterous hands, where tactile feedback can enable even richer and more dexterous manipulation skills.

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Supplementary Materials

Contents

1	Introduction	1
2	Related Works	2
3	Visuo-Tactile Data Collection System	3
3.1	Scalable Flexible Tactile Sensors	3
3.2	Portable Multi-Modal Sensing System	4
4	Visuo-Tactile Representation and Policy Learning	4
4.1	Problem Formulation	4
4.2	Stage 1: Visuo-Tactile Representation Learning	5
4.3	Stage 2: Policy Learning via Behavior Cloning	6
5	Experiments	6
5.1	Large-Scale Visuo-Tactile Data and Pre-Training	7
5.2	Experiments Setup	7
5.3	Analysis and Discussion	11
6	Conclusion and Limitations	12
A	Downstream Task Training Analyses	17
B	Baseline Failure Analysis	17
C	Training Hyperparameters for pretraining	19
D	Detailed Hyperparameters for downstream task training	20
E	Extended Task Descriptions	20
E.1	Details for the Test Tube Collection Task	20
E.2	Details for the Fluid Transfer Task	21
E.3	Details for the Erase Whiteboard Task	21
E.4	Details for the Pencil Insertion Task	22

A Downstream Task Training Analyses

To evaluate the impact of visuo-tactile pretraining on downstream task performance, we analyze both task success rates and training convergence behavior. In addition to final success rates for the Test Tube Collection task under low-data and low-epoch regimes with and without pretraining (see Section V in main paper), we also track training loss curves for two representative tasks—Test Tube Collection and Pencil Insertion—with and without pretraining, as shown in Figure 10.

We plot training loss from epoch 5 to epoch 60 for both settings. Across both tasks, we observe that policies with visuo-tactile pretraining consistently achieve lower loss values throughout training compared to those without pretraining. This indicates faster convergence and improved training stability.

Importantly, even when policies trained with and without pretraining converge to similar loss levels during training, we observe that policies with pretraining consistently achieve higher success rates in real-world rollouts. We attribute this discrepancy to the difference between open-loop training (measured by loss) and closed-loop execution (evaluated via physical rollouts). In real robot experiments, downstream tasks are subject to unpredictable variations in object positions, environmental dynamics, and contact conditions. Policies with visuo-tactile pretraining are more robust to these variations because they have learned to rely on tactile cues that generalize across scenarios. As a result, they exhibit more reliable and consistent behavior—particularly in out-of-distribution conditions.

B Baseline Failure Analysis

To better understand the challenges involved in fine-grained real-world manipulation, we analyze common failure modes across baseline policies for the four core tasks. Figure 11 presents representa-

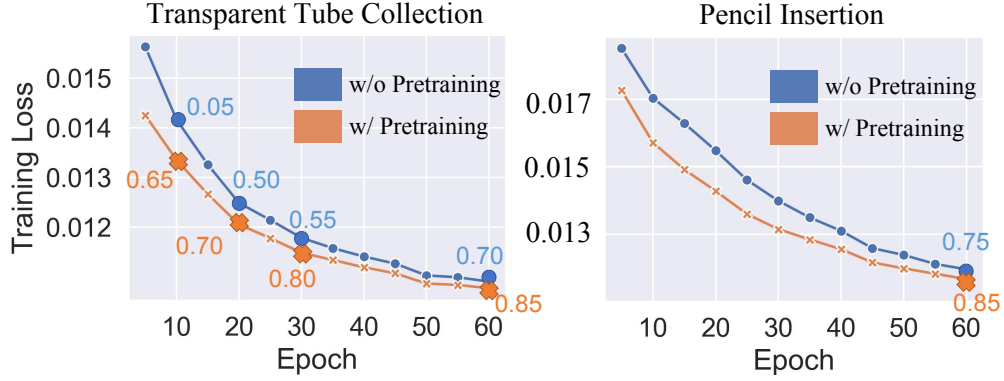


Figure 10: **Downstream policy training loss for two tasks.** Training loss curves from epoch 5 to 60 for Test Tube Collection and Pencil Insertion tasks. Numbers along each curve indicate rollout success rates at corresponding epochs. Policies with pretraining show faster convergence and consistently higher success rates compared to policies without pretraining.

tive examples of failures for both vision-only policies and policies without cross-attention. These cases highlight the limitations of relying solely on visual or static tactile input and emphasize the importance of integrated visuo-tactile feedback.

(a) Test Tube Collection In the test tube collection task, vision-only policies often struggle to differentiate between the reorientation and insertion phases. As shown in Figure 11(a), a typical failure involves prolonged hesitation when deciding whether to continue reorienting or to proceed with insertion. Even after a successful reorientation, the policy may continue the motion unnecessarily, suggesting uncertainty about the current phase of the task.

Although policy without cross-attention transitions more decisively between phases, it can become overly dependent on localized tactile cues. This sometimes results in the robot getting stuck during reorientation, where it maintains excessive contact without progressing. Additionally, the policy without cross-attention frequently fails during insertion, likely due to insufficient use of visual input for aligning the test tube with the narrow slots of the test tube rack.

(b) Fluid Transfer Fluid transfer requires precise, force-sensitive control—something vision-only policies typically struggle to provide. As illustrated in Figure 11(b), a common failure during fluid extraction involves excessive squeezing, where the robot either applies too much pressure or continues to squeeze for too long. Without tactile feedback, the policy relies entirely on ambiguous visual cues, often resulting in drawing too much fluid. This overfilling can hinder subsequent steps, particularly during the expelling phase, where the robot may fail to fully release the fluid. The result is an imprecise release and overall task failure.

Failures are also frequent during the expelling phase. The policy may skip the expel fluid step entirely, particularly when the fluid is not clearly visible—either due to an insufficient amount drawn or when the second beaker contains nearly no water. These behaviors suggest that the vision-only policy is overly sensitive to subtle environmental variations and fails to detect critical task-related cues.

(c) Erase Whiteboard Erasing a whiteboard requires consistent force against a flat surface. Vision-only policies often fail because they rely on incomplete visual feedback on the contact of the eraser with the board. As a result, insufficient pressure is applied, leaving visible marks and leading to task failure (Figure 11(c)).

On the other hand, policies without cross attention tend to overcompensate by pressing too hard in an attempt to maximize tactile feedback. This often results in exceeding force thresholds, triggering safety stops, and prematurely terminating the task.

(d) Pencil Insertion Pencil insertion requires precise in-hand reorientation before the final insertion step. Vision-only policies often under-rotate the pencil, especially when it is held low in the gripper

Typical Failure Cases of Baselines

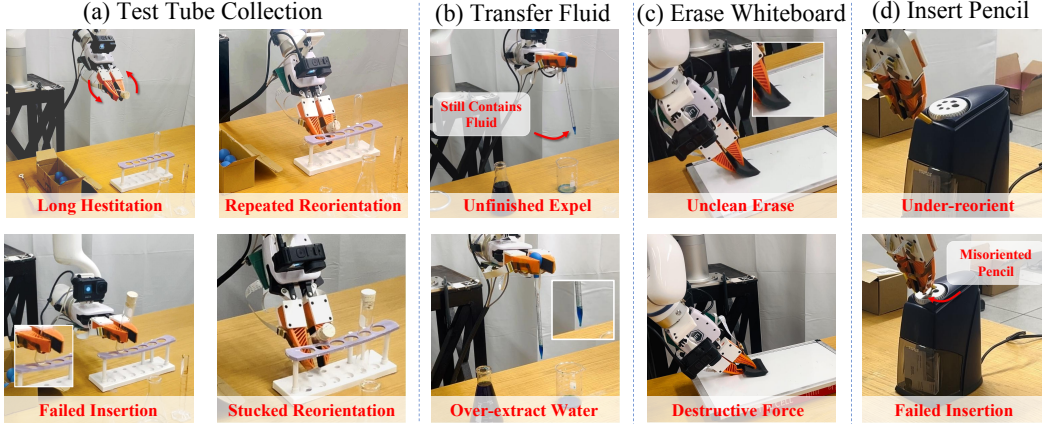


Figure 11: **Failure Cases** We present typical failure cases of the baseline method for all four tasks and analyze the reasons for these failures to highlight the complexity of the tasks and the importance of tactile feedback and pretraining during these steps.

and partially occluded from view, or when the pencil is only slightly inclined (Figure 11(d)). When the pencil is nearly parallel to the gripper, it appears very small in the visual frame, further reducing the effectiveness of visual feedback. Lacking accurate information about the in-hand state, the policy frequently misjudges the pencil’s orientation, leading to failed insertions.

C Training Hyperparameters for pretraining

Table 2: Pretraining hyperparameters and training configuration

Parameter	Value
Seed	42
Tactile embed dim	768
Tactile patch size	4
Number of attention heads	8
Cross-attention dropout	0.20
Decoder hidden dim	768
Batch size	128
Epochs	5
CLIP learning rate	3×10^{-5}
Encoder learning rate	1×10^{-4}
Weight decay	2×10^{-3}
Warmup ratio	0.10
Scheduler	Linear warmup + Cosine annealing
Optimizer	AdamW
Validation ratio	0.10
Tactile masking ratio	60%–80% (random patch masking during pretraining)

D Detailed Hyperparameters for downstream task training

Table 3: Training hyperparameters for policy learning

Parameter	Value
Observation Settings	
Image observation horizon	2
Proprioception observation horizon	2
Action horizon	16
Observation resolution	224×224
Optimization Parameters	
Optimizer	AdamW
Optimizer momentum	$\beta_1 = 0.95, \beta_2 = 0.999$
Learning rate (action diffusion)	3×10^{-4}
Learning rate (visual encoder)	3×10^{-5}
Learning rate schedule	Cosine decay
Batch size	64
Train diffusion steps	50
Inference denoising steps	16

E Extended Task Descriptions

In this section, we discuss the detailed information of the four tasks described in the main paper. We will discuss the motions and evaluation metrics for each step, and highlight how these steps demonstrate the capabilities of our tactile sensors. The typical failure cases are shown in Fig. 11 and will be discussed in the following sections.

E.1 Details for the Test Tube Collection Task

Initial Positions. At the start of each episode, the robot is positioned near a cluttered box containing a test tube. The test tube is partially visible, with its close end typically protruding slightly outside the box, while the rest is embedded among other objects. Several aspects are randomized across episodes: the robot’s initial pose, the orientation and depth of the test tube, the degree of surrounding clutter, and the color of the wooden cork. These variations introduce uncertainty in the grasping setup and require precise spatial reasoning and tactile feedback for reliable performance.

Step 1: Grasp the Test Tube. The robot reaches into the box to grasp the partially exposed test tube. Depending on its orientation and surrounding clutter, the robot may need to adapt to a wide range of grasping conditions. This step demonstrates the strength of our thin, flexible tactile sensor in enabling precise manipulation within narrow and constrained environments. *Evaluation Metrics:* The task is considered successful if the robot securely grasps the test tube and returns to a neutral position to proceed to the next step.

Step 2: Reorient the Test Tube. After grasping, the robot moves to the test tube rack and uses its edge to rotate the test tube by approximately 70 degrees. This reorientation is particularly challenging due to the test tube’s transparency, which can cause vision-only policies to misjudge in-hand orientation and either under- or over-rotate. In contrast, tactile feedback provides clear contact signals, enabling more reliable reorientation. *Evaluation Metrics:* The task is successful if the robot completes a single, correct reorientation. The task fails if multiple attempts are required or if the final orientation remains incorrect.

Step 3: Insert the Test Tube. Finally, the robot attempts to insert the reoriented test tube into a tight-fitting slot in the rack. Given the transparent material and narrow tolerance, accurate alignment is critical. Tactile sensing plays a key role in localizing the test tube body’s position and guiding the

insertion process. *Evaluation Metrics:* The task is considered successful if the test tube is inserted fully and securely into the rack without collision or misalignment.

E.2 Details for the Fluid Transfer Task

Initial Positions. The robot begins with its parallel grippers positioned around the bulb of a pipette, roughly centered between its fingers. Across episodes, several variables are randomized: the robot’s initial pose, the orientation and position of the pipette within the first beaker, the location of both beakers, the amount of water in each beaker, and the water’s color and opacity. These variations introduce uncertainty in grasping and perception, making tactile sensing essential for accurately locating the bulb and applying the correct amount of force.

Step 1: Grasp and Lift the Pipette. The robot first applies gentle pressure to the pipette’s bulb to extract water from the first beaker. This step requires fine control of squeezing force—too little and not enough water is drawn, too much and the pipette may slip or spill. The robot must also detect when a sufficient amount of liquid has been collected, and then lift the pipette while maintaining a stable grasp, preparing for the transfer. Visual feedback is unreliable here due to partial occlusion of the bulb and the small volume of liquid involved. Our tactile sensors provide accurate contact localization and force feedback, enabling the robot to perform this motion precisely and without spillage. *Evaluation Metrics:* The task is considered successful if the robot extracts an appropriate amount of liquid and lifts the pipette without spilling.

Step 2: Transfer to the Second Beaker. With the pipette securely grasped and filled, the robot moves toward the second beaker. It must maintain consistent pressure on the bulb throughout the motion to avoid accidental release or dropping the pipette. This step requires precise motion planning and continuous force regulation, as small deviations in grip or trajectory can lead to fluid loss. Tactile sensing ensures grip stability, while vision assists in coarse positioning. *Evaluation Metrics:* The task is successful if the robot moves the pipette directly above the second beaker without spilling any water during the transfer.

Step 3: Expel Fluid. Once positioned above the second beaker, the robot lowers the pipette into the beaker and squeezes the bulb to release the water. This step requires detecting the correct contact point and applying the appropriate force to fully expel the liquid. Vision-only policies often struggle here due to the transparency of the pipette and fluid, making it difficult to determine whether expulsion has occurred. In contrast, our visuo-tactile policy uses force signals to monitor the release and avoid lifting prematurely. *Evaluation Metrics:* The task is considered successful if all liquid is expelled into the second beaker and the pipette is lifted cleanly, with no water remaining or unintended spillage.

E.3 Details for the Erase Whiteboard Task

Initial Positions. The robot starts with a soft foam brush held vertically in its gripper. Across episodes, several aspects of the setup are randomized: the robot’s initial pose, the height and position of the whiteboard, and the exact location of the brush within the gripper. These variations introduce uncertainty in alignment and contact, making tactile sensing critical for successful execution.

Step 1: Erase First Stroke. The robot begins with the foam brush held vertically in its gripper. To prepare for erasing, it first rotates its gripper by approximately 90 degrees, reorienting the brush into a horizontal position. It then moves toward the whiteboard and attempts to erase the letter “T” using a vertical stroke. The robot must apply a slight downward force and rely on tactile feedback to confirm contact with the whiteboard surface. This step is challenging because visual input alone may not reliably indicate whether the brush has made effective contact, especially in the presence of occlusions or slight misalignments. *Evaluation Metrics:* The task is considered successful if the robot fully erases the letter “T” with proper contact and without slipping or losing control of the brush.

Step 2: Erase Second Stroke. After completing the vertical stroke, the robot reorients the brush back to a vertical configuration. It then performs a horizontal stroke to erase the letters “ac.” Tactile sensing is again crucial—not only for confirming contact, but also for triggering the transition from motion to

active erasure. *Evaluation Metrics:* The task is considered successful if the robot cleanly erases the letters "ac" while maintaining stable and continuous control of the brush.

E.4 Details for the Pencil Insertion Task

Initial Positions. The robot begins with each episode holding a slightly inclined pencil in its gripper. Several factors are randomized across episodes: the pencil's orientation and position within the gripper, the placement and height of the pencil sharpener, the size and alignment of the insertion hole, and the robot's initial pose. These variations introduce uncertainty that requires fine-grained control and perception.

Step 1: Reorient the Pencil. The robot first moves toward the vertical surface of the pencil sharpener and uses it as a reference to reorient the pencil. The goal is to align the pencil so that it becomes parallel to the gripper and ready for insertion. This step is challenging due to the pencil's small size and partial occlusion within the gripper, which makes it difficult to estimate its pose using vision alone. Tactile feedback provides critical information for detecting contact and adjusting alignment. *Evaluation Metrics:* This step is considered successful if the pencil is reoriented such that it lies parallel to the robot's gripper, preparing it for precise insertion.

Step 2: Insert the Pencil. With the pencil correctly oriented, the robot must now insert it into the top hole of the pencil sharpener. This requires accurate positioning, controlled force, and fine alignment to avoid missing the hole or applying too much pressure, which could cause the pencil to drop or get stuck. Tactile sensing ensures that the robot detects the hole and regulates the insertion motion appropriately. *Evaluation Metrics:* The task is considered successful if the pencil is inserted smoothly into the sharpener without slipping, jamming, or requiring realignment.